Procedure for Resolving Huawei Access Node Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) at Jayanagar Metro Site

Node: JAY-ACC-60 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6552

Description

The optical transport failure experienced on the Huawei access node (Role: Access, Node ID: JAY-ACC-60, Site Location: Jayanagar Metro Site) can be attributed to the degradation of SFP/XFP modules or physical damage to the fiber optic cables, resulting in a decrease in receive (Rx) or transmit (Tx) power. This failure domain is specific to the access node due to its role in terminating the fiber optic connections and its reliance on the SFP/XFP modules for data transmission. The Huawei equipment in question uses a proprietary protocol to manage and control the SFP/XFP modules, which can lead to issues with Rx/Tx power levels if not properly configured or if the modules are degraded. Furthermore, environmental factors such as temperature fluctuations, humidity, or physical stress can also contribute to the degradation of the SFP/XFP modules or fiber optic cables.

Pre-requisite Checks

- Verify that the site is accessible and the power is on.
- Check the redundant link status to ensure it is active and functioning correctly.
- Confirm that the issue is specific to the Node ID: JAY-ACC-60 and not a site-wide problem.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP module	Medium
Fiber optic cable damage	Low
Rx/Tx power level misconfiguration	High

Diagnostic Commands & Expected Outputs

```
Command: display sfp info, Expected Output: SFP information, including SFP status, model number, and serial number, Description: This command is used to display information about the SFP/XFP modules, including their status, model number, and serial number. This information can be used to identify potential issues with the modules.

Command: display interface, Expected Output: Interface information, including Rx/Tx power levels, Description: This command is used to display information about the interfaces, including the receive and transmit power levels. This information can be used to identify potential issues with Rx/Tx power levels.
```

```
Command: display fiber status, Expected Output: Fiber status
information, including fiber type and length, Description: This
command is used to display information about the fiber optic cables,
including the type and length of the fiber. This information can be
used to identify potential issues with the fiber optic cables.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP module, Command: undo sfp replace , Description: This command is used to replace the degraded SFP/XFP module with a new one. The parameter specifies the slot where the new module will be installed.

Step 2: Step: Adjust the Rx/Tx power level, Command: set interface power , Description: This command is used to adjust the Rx/Tx power level on the specified interface. The parameter specifies the interface where the power level will be adjusted, and the parameter specifies the new power level.

Step 3: Step: Inspect and repair the fiber optic cable, Command: display fiber status, Description: This command is used to display information about the fiber optic cables, including the type and length of the fiber. This information can be used to identify potential issues with the fiber optic cables and determine the necessary repairs.

Post-Check Verification

```
Command: display sfp info, Expected Output: SFP information, including
SFP status, model number, and serial number, Description: This command
is used to verify that the SFP/XFP module has been replaced or
repaired successfully.

Command: display interface, Expected Output: Interface information,
including Rx/Tx power levels, Description: This command is used to
verify that the Rx/Tx power levels have been adjusted correctly.

Command: display fiber status, Expected Output: Fiber status
information, including fiber type and length, Description: This
command is used to verify that the fiber optic cable has been
inspected and repaired successfully.
```

Roll-back Procedures

- Step: Revert the SFP/XFP module replacement, Command: undo sfp replace , Description: This command is used to revert the replacement of the SFP/XFP module if the new module is not functional.

- Step: Revert the Rx/Tx power level adjustment, Command: undo set interface power , Description: This command is used to revert the adjustment of the Rx/Tx power level if it was not set correctly.

- Step: Revert the fiber optic cable repair, Command: undo display fiber status, Description: This command is used to revert the inspection and repair of the fiber optic cable if it was not necessary.